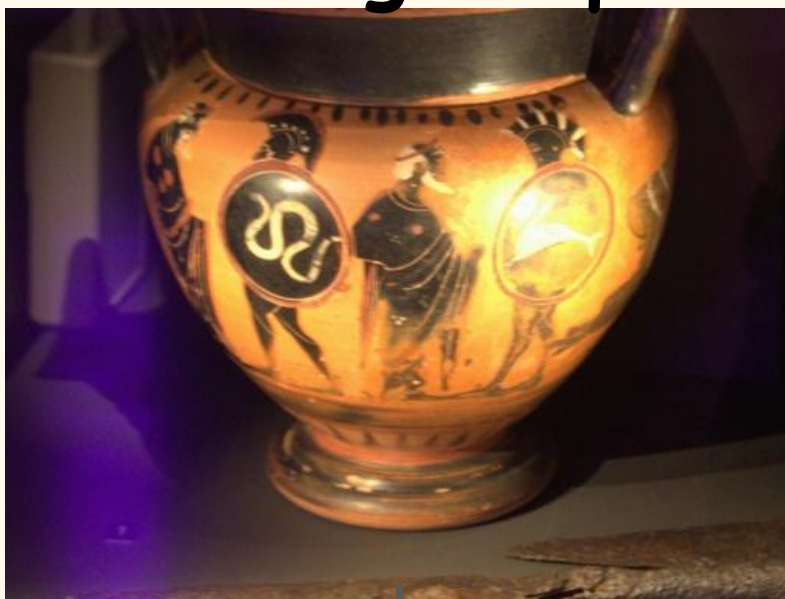


Prompt Input

Assess the overall perceptual quality. Respond with exactly containing: "reasoning" and "rating". "reasoning" contains: "evidence" and "solution".
"Evidence": visible image evidence that determines its current overall perceptual quality and location.
"Solution": a coherent image-edit plan that causally addresses the evidence. Retain the same semantic meaning as the input.
"Rating": a numeric string from 1.00 to 5.00. "1.00" represents the worst quality and "5.00" represents excellent quality.

Image Input



Actor



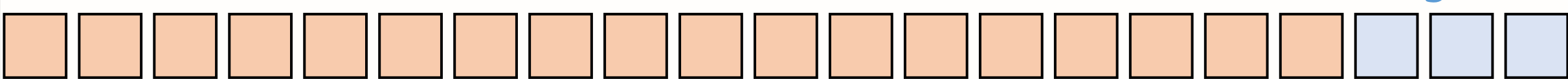
Vision Encoder

Aligner

LLM

Reasoning Tokens: Evidence, Solution

Rating Tokens



Original Image Reasoning Tokens

Editor



Vision Encoder

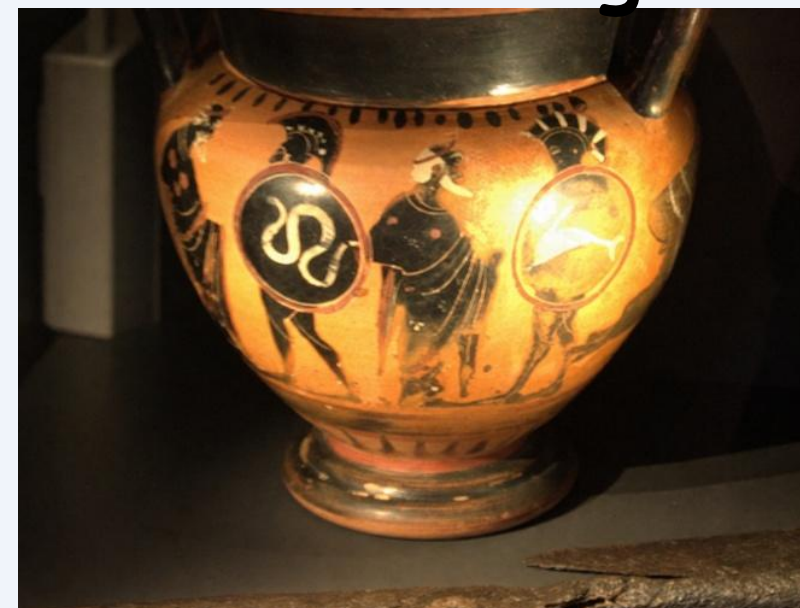
Text Encoder

Editing Model

Edited Image 1



Edited Image k



Edited Image K



Judge



Pretrained MLLM

Original and Edited Images

Reasoning: The vase is clearly recognizable, but strong purple lens flare, overexposed highlights, moderate softness

Rating: 3.20

Judge Original

Reasoning: The subject is clear and reasonably sharp, but strong highlights on the vase are slightly overexposed.

Rating: 3.70

Judge k: $\Delta_s = 0.50$

Reasoning: Although lighting is uneven, the painted designs are discernible and draw the attention.

Rating: 3.30

Judge 1: $\Delta_s = 0.10$

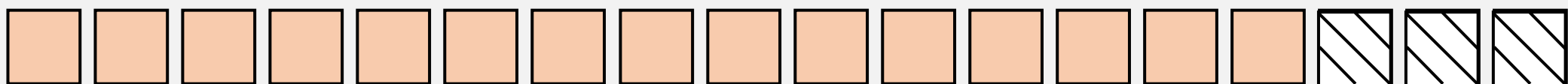
Reasoning: The vase is clear and detailed, though uneven lighting and slight softness reduce the overall quality.

Rating: 4.25

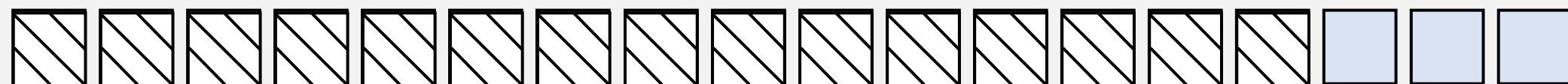
Judge K: $\Delta_s = 1.05$

Masked Credit Assignment

Reasoning: greater quality gains, higher credit.



Rating: better margin estimation, higher credit.



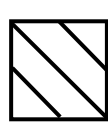
Optimization Target: (Rollout K's Reasoning, Rollout k's rating)



trainable



frozen



masked token